

# ECN 594: Midterm Exam

January 18, 2026

## Instructions:

- You have **80 minutes**
- You may bring a calculator and notes on a two-sided cheat-sheet (letter-size paper)
- Please be neat. If your work is too messy it will not be graded.
- Be sure to show your working.
- This is a long exam, so there are lots of ways to get points. If you get stuck, move on!
- Good luck!

Name: \_\_\_\_\_

|           |    |    |    |    |       |
|-----------|----|----|----|----|-------|
| Question: | 1  | 2  | 3  | 4  | Total |
| Points:   | 30 | 30 | 20 | 20 | 100   |
| Score:    |    |    |    |    |       |

## 1. Short Answer Questions (30 points)

1. For each question, give a number, True/False/NEI, or a 1-2 sentence answer.
  - (a) (4 points) A monopolist faces demand with elasticity  $\varepsilon = -2$  and has  $MC = \$30$ . What is the profit-maximizing price?
  - (b) (4 points) Consider the logit utility:  $u_{ijt} = \beta \cdot \text{sugar}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$ . What is the economic interpretation of  $\alpha$ ?
  - (c) (4 points) True, False, or NEI: If we add 5 to the mean utility of all goods (including the outside option), the market shares do not change.
  - (d) (4 points) Why are BLP instruments (characteristics of other products) *relevant* for price? (1-2 sentences)
  - (e) (4 points) True, False, or NEI: In the logit model with small market shares, higher-priced products typically have more elastic demand.
  - (f) (3 points) True, False, or NEI: Under average cost pricing ( $P = AC$ ), deadweight loss is zero.
  - (g) (4 points) A researcher claims that consumer surplus in logit is  $CS = \frac{1}{|\alpha|} (\exp(\delta_1) + \exp(\delta_2))$ . Consider a market with two goods ( $\delta_1 = \delta_2 = 2$ ) and an outside option. Is this formula correct?

(h) (3 points) True, False, or NEI: Under perfect price discrimination, deadweight loss is zero.

## 2. Demand Estimation (30 points)

2. A researcher wants to estimate demand for automobiles using a logit model:

$$u_{ijt} = \beta \cdot \text{mpg}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

where  $\text{mpg}_j$  is fuel efficiency,  $p_{jt}$  is price,  $\xi_{jt}$  is unobserved product quality, and  $\varepsilon_{ijt}$  is i.i.d. Type 1 Extreme Value.

- (a) (6 points) Define the mean utility  $\delta_{jt}$ . Using the Berry inversion, derive an expression for  $\delta_{jt}$  in terms of observed market shares  $s_{jt}$  and the outside option share  $s_{0t}$ .
- (b) (6 points) Write down the linear regression equation the researcher would estimate to recover  $\beta$  and  $\alpha$ . Clearly identify the dependent variable, regressors, and error term.
- (c) (6 points) A researcher proposes using “Hausman instruments” (prices of the same car in other markets) to address the endogeneity problem. Explain one reason why these instruments might be problematic.

- (d) (6 points) Compared to the basic logit, what is one advantage of adding demographic interactions (e.g.,  $\text{income} \times \text{price}$ ) to the model?
- (e) (6 points) A researcher takes all Toyota Corollas and splits them into two “products” based on the last digit of their license plate: “Corolla-Odd” and “Corolla-Even.” What will the logit model predict about the change in consumer surplus from this split? Should we believe this prediction? Explain.

### 3. Two-Part Tariffs (20 points)

3. A gym charges a monthly membership fee  $F$  and a per-visit price  $p$ . Marginal cost per visit is  $c = 2$ .

#### Part A: Homogeneous consumers

There are 100 identical consumers, each with demand  $q = 20 - 2p$  for gym visits per month.

- (a) (5 points) Write down the gym's profit as a function of  $F$  and  $p$ . (Hint: What is consumer surplus at price  $p$ ? What is the maximum  $F$  the gym can charge?)

- (b) (5 points) Show that the profit-maximizing per-visit price is  $p^* = c = 2$ . (Hint: Take the derivative of profit with respect to  $p$  and set it equal to zero.)

**Part B: Heterogeneous consumers**

Now suppose there are 50 “heavy users” with demand  $q_H = 10 - p$  and 100 “light users” with demand  $q_L = 5 - p$ . Marginal cost is  $c = 0$ .

(c) (10 points) The gym is considering two pricing strategies:

(a)  $F = 50, p = 0$

(b)  $F = 12.5, p = 0$

Calculate profit under each strategy. Which is better and why?

#### 4. Self-Selection (20 points)

4. A software company sells two versions: Pro (P) and Basic (B). There are two types of users: businesses with high willingness to pay and students with low willingness to pay.

| Consumer Type       | Willingness to Pay |               |
|---------------------|--------------------|---------------|
|                     | Pro Version        | Basic Version |
| Business (40 users) | \$400              | \$150         |
| Student (60 users)  | \$120              | \$100         |

Marginal cost is \$30 for both versions.

- (a) (6 points) If the company could perfectly identify consumer types, what prices would it charge for each version to each type? What is total profit?
- (b) (6 points) The company cannot identify types but offers both versions. Write down the incentive compatibility (IC) constraint that ensures businesses buy Pro rather than Basic.

- (c) (8 points) Find the profit-maximizing prices for Pro and Basic under self-selection. (Hint: Which IC constraint binds? Set it as an equality and solve.)